

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The hydrosphere: hydrological cycle & unsustainable exploitation - Set 1

Mark scheme

AQA spec 3.2.2.1-3.2.2.3; The physical environment; textbook pp. 114-121

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $13 / 480 = 0.0271$ years 1 0.0271×365 1 = 10 days (accept 9.9-10) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (days) is already printed on the answer line. • residence time → the time a molecule stays in the atmosphere / atmospheric lifetime Do not credit • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Question	Marking guidance	Marks
01.2	Calculation 1 mark for the correct answer; working does not have to be shown. 1 $1\,1340\,000 / 425 = 3153$ years (accept 3100-3200) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • residence time → the time a molecule stays in the atmosphere / atmospheric lifetime Do not credit • A bare wrong answer with no working.	1
01.3	1 mark each 1 mark for each of the 2 points below. 1 It has a very short residence time / very rapid transfer rate ($480 \times 10^{15} \text{ dm}^3$ per year) 1 So it moves large volumes of water each year, delivering precipitation to land – small reservoirs with fast transfers respond quickly to change Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • residence time → the time a molecule stays in the atmosphere / atmospheric lifetime Do not credit • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 The ocean is so large that an added or removed volume is a tiny proportion of the total 1 The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • residence time → the time a molecule stays in the atmosphere / atmospheric lifetime Do not credit • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Question	Marking guidance	Marks
01.5	1 mark each 1 mark for each of the 2 points below. 1 Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) 1 But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infiltration → water soaking into the soil surface transpiration → water lost as vapour from the leaves Do not credit - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	
02.1	2 marks per point 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. 2 Deforestation: less interception so less evaporation from leaves - less transpiration so less water returned to the atmosphere and less precipitation downwind - more infiltration and surface runoff, higher river flows / flooding 2 Urban development: impermeable concrete / tarmac reduce infiltration and aquifer recharge - runoff is faster and greater, increasing river flooding downstream - drains carry water quickly to rivers 2 Agriculture: soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff - irrigation increases evaporation - loss of soil biota (worms) reduces infiltration and water retention - drainage lowers water tables Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. transpiration → water lost as vapour from the leaves infiltration → water soaking into the soil surface impermeable → does not let water through Do not credit 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	6

Question	Marking guidance	Marks
02.2	Any 2 of 4 1 mark each for any 2 of the 4 points below, in any order, maximum 2. 1 Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation 1 More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods 1 Changed wind patterns move humid air to different areas 1 Sea level rise / salt incursion into coastal aquifers Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
02.3	1 mark each 1 mark for each of the 2 points below. 1 Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity 1 Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
03.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 $(68\,000 - 7\,000) / 68\,000 \times 100$ 1 = 89.7% (accept 90%) Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (%) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	2
03.2	1 mark each 1 mark for each of the 2 points below. 1 Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued 1 Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
03.3	1 mark each 1 mark for each of the 4 points below. 1 Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress – pollution by pesticides, fertilisers and industrial waste concentrated in the lake 1 Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) 1 Loss of wetland habitats and wildlife – local climate more extreme (loss of the moderating effect of the water body) 1 Loss of livelihoods – conflict between the countries sharing the rivers Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. range of tolerance → the range of conditions in which a species can survive salinisation → salt building up in the soil / soil becoming salty Do not credit - pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
03.4	1 mark each 1 mark for each of the 2 points below. 1 Upstream abstraction for irrigation reduces the flow available downstream 1 Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream – growing populations increase demand Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. abstraction → water taken out of a river, lake or aquifer HEP → electricity generated from falling water Do not credit - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
04.1	Calculation 1 mark for the correct answer; working does not have to be shown. 1 $(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$ Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (m yr^{-1}) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	1

Question	Marking guidance	Marks
04.2	1 mark each 1 mark for each of the 2 points below. 1 Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation 1 The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Abstraction → water taken out of a river, lake or aquifer infiltration → water soaking into the soil surface percolation → water moving downwards through soil or rock	2
04.3	1 mark each 1 mark for each of the 3 points below. 1 Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) 1 Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) 1 Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. aquifer → a permeable rock that stores groundwater / underground water store Permeability → how easily water passes through / lets water through Impermeable → does not let water through	3
04.4	Any 4 of 5 1 mark each for any 4 of the 5 points below, in any order, maximum 4. 1 Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) 1 Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) 1 Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds 1 Ecological impacts: plants needing high water availability die – wetland animals lost – food chain effects 1 Reduced supplies: abstraction may suddenly have to stop after years of hidden decline Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. aquifer → a permeable rock that stores groundwater / underground water store abstraction → water taken out of a river, lake or aquifer Do not credit - Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
05.1	1 mark each 1 mark for each of the 3 points below. 1 The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter 1 Current recharge is little or none so any abstraction reduces the stored volume 1 The large volume means shortages take a long time to appear, but eventually the supply runs out – ancient water is often saline, causing soil salinisation Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • abstraction → water taken out of a river, lake or aquifer • aquifer → a permeable rock that stores groundwater / underground water store • sustainable → meeting present needs without stopping future generations meeting theirs Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.2	1 mark each 1 mark for each of the 3 points below. 1 Two satellites whose orbits are affected by the force of gravity 1 The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass 1 Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • aquifer → a permeable rock that stores groundwater / underground water store	3
05.3	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Low-pressure irrigation pipes to reduce leaks 1 Growing crops that need less water than wheat / maize 1 Charging consumers by volume used (metering) 1 Inter-basin transfer of water 450 km from the Yangtze Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • aquifer → a permeable rock that stores groundwater / underground water store Do not credit • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Question	Marking guidance	Marks
05.4	1 mark for the point below. 1 Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. aquifer → a permeable rock that stores groundwater / underground water store Do not credit 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
06.1	Marks shared Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. up to 2 Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) up to 2 Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river up to 2 Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds up to 2 Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion up to 2 Microclimate: water's high heat capacity reduces temperature fluctuations - higher wind speeds - more evaporation, humidity, cloud and precipitation downwind Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
06.2	1 mark each 1 mark for each of the 2 points below. 1 Vegetation / dead organic matter under water decays anaerobically 1 Releasing methane, a powerful greenhouse gas Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Question	Marking guidance	Marks
06.3	1 mark each 1 mark for each of the 2 points below. 1 After heavy rain surplus water is stored, reducing peak flow / flooding downstream 1 In dry periods the dam releases water to maintain river levels / supplies Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	